Project Progress Report and Next-Review Plan

Implementation and Validation of a Simple-Structure Constrained PD-Type Attitude Controller for Quadrotor Simulation

Group AB02 · Introduction to Drones (22AIE448) · Amrita Vishwa Vidyapeetham

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Base paper	M. Golestani, K. A. Alattas, S. Ud Din, S. Mobayen, A. K. Alanazi, A. Fekih, "A Simple Structure Constrained Attitude Control for Rigid Bodies: A PD-Type Control," *IEEE Access*, vol. 10, pp. 10202–10209, 2022. DOI 10.1109/ACCESS.2022.3144627
Repository	Drones_S5_AB02_PD_Control
Contents	Summary · Paper formulation · Work completed · Reproduction results · Equation audit · Gazebo status · Next-review plan · Claims register

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0. Summary

The base paper proposes a **constrained PD-type attitude controller** that keeps both the attitude quaternion and the angular velocity inside time-varying, finite-time prescribed-performance envelopes, under actuator saturation and bounded disturbances. Its main selling point is *simplicity*: a normal PD law plus one barrier-weighted term and one scalar adaptive estimate, instead of the transformed-error machinery of classical prescribed-performance control (PPC).

This report states (i) what has been implemented so far, (ii) what has been **verified** against the paper's equations, (iii) what is **not yet working**, and (iv) exactly what will be implemented and demonstrated at the next review.

Status at a glance

Work item	Status	Evidence
Paper equations (1)–(14) implemented as a direct ODE simulation (MuJoCo notebooks)	**Done**	`Drones_Mujoco_Pointing_Trial.ipynb`
Interactive browser simulator (RK4, constrained vs PD, live plots)	**Done** (equation form to be aligned, §5)	`index.html`, `app.js`, `simulation.js`
Faithful numerical reproduction of paper Part 1 and Part 2 with quantitative metrics	**Done** (this report)	`paper_reproduction.py`, `docs/results/metrics.json`
Equation-to-code audit across all implementations	**Done** (this report, §5)	Table 5.1, findings F1–F8
MuJoCo attitude-pointing and point-to-point flight demos	**Done** (simplified force/torque actuators)	`Drones_Mujoco_*.ipynb`
ROS 2 + Gazebo X3 pipeline (pose feedback, motor-speed commands, logging, plotting)	**Done**	`gazebo.py`
Sustained hover on the Gazebo X3 with the paper controller	**In progress**	The vehicle takes off and climbs, but attitude control is not maintained and the run ends in a crash; the cause is under investigation
Disturbance-injection and constrained-vs-PD benchmark **on the X3**	**Planned** (next review)	flags exist in `gazebo.py`, not yet run
Stable square-trajectory baseline (Gazebo built-in velocity controller)	**Done, but separate from the paper controller**	commit script, CSV and plots to the repository before the review

> **Scope statement.** The stable square-trajectory demo uses Gazebo's built-in low-level controller, so it validates the simulation and evaluation pipeline, **not** the paper's controller. The MuJoCo point-to-point flights use direct force/torque actuators, not four-rotor mixing. Neither is presented here as evidence that the paper's controller flies the X3.

1. Problem and Contribution of the Base Paper

Problem. Attitude control of a rigid body (spacecraft or UAV) must achieve (a) a prescribed transient and steady-state behaviour for *both* the attitude and the angular velocity, (b) despite actuator saturation and an unknown bounded disturbance.

Why it is hard. Conventional PPC transforms each constrained error through a function such as $S(\varepsilon) = \tanh(\varepsilon)$ and designs the controller on the transformed variable. This yields partial-derivative terms and strongly coupled dynamics, and the complexity roughly doubles when the angular velocity is constrained too.

Paper's idea. Do not control the transformed variable. Instead:

1. Keep a **conventional PD law** to drive the states to zero.
2. Add an **auxiliary term** u_{cons} whose gain grows as a state approaches its envelope, pushing it back.
3. Add a scalar **adaptive estimate** $\hat{\theta}$ of the lumped disturbance-plus-saturation bound.

4. Use a **finite-time PPF (FTPPF)** so the envelope reaches its final value ρ_T exactly at a chosen settling time T_s.

Claimed advantages. Simple structure; overshoot, convergence rate and ultimate bound set by the FTPPF parameters rather than by the gains; behaviour under saturation; better performance than conventional PD at similar peak torque.

2. Mathematical Formulation (as in the paper)

Equation numbers follow the paper so the professor can cross-check directly.

2.1 Plant: rigid-body attitude with saturation (paper Eq. 1, 3, 4, 5)

With unit quaternion $q = [q_v^\top,\, q_4]^\top$, $q^\top q = 1$, angular velocity $\omega \in \mathbb{R}^3$ in the body frame, inertia $J \succ 0$, control torque u, disturbance d with $\|d\| \le \bar{d}$:

$$\dot{q}_v = \tfrac{1}{2}\,(q_4 I_3 + q_v^\times)\,\omega, \qquad \dot{q}_4 = -\tfrac{1}{2}\,q_v^\top \omega, \qquad J\dot{\omega} = -\omega^\times J\omega + u + d$$

where $z^\times$ is the skew-symmetric matrix of z. Actuator saturation (Eq. 3–4) is modelled as $\mathrm{sat}(u_i) = u_i + \pi_i$, with $\pi_i = 0$ if $|u_i| < u_{\max,i}$ and $\pi_i = \mathrm{sgn}(u_i)\,u_{\max,i} - u_i$ otherwise. Collecting the unknown bounded terms $T_d = d + \Pi$ with $\|T_d\| \le \theta$ (unknown constant):

$$J\dot{\omega} = -\omega^\times J\omega + u + T_d \qquad \text{(Eq. 5)}$$

2.2 Prescribed performance functions (Eq. 6–9)

The constraint to enforce, for $i = 1, 2, 3$:

$$-\rho_q(t) < q_i(t) < \rho_q(t), \qquad -\rho_\omega(t) < \omega_i(t) < \rho_\omega(t)$$

The classical exponential PPF (Eq. 7) is $\rho_q(t) = (\rho_{q0} - \rho_{q\infty})e^{-\kappa_q t} + \rho_{q\infty}$, which only *approaches* its limit. The paper's **finite-time PPF** (Eq. 8) reaches it at T_s:

$$\rho(t) = \begin{cases} \left(\rho_0 - \rho_T\left(1 + \dfrac{t}{T_s}\right)\right)\exp\left(\dfrac{\kappa\, t}{t - T_s}\right) + \rho_T, & 0 \le t < T_s \\ \rho_T, & t \ge T_s \end{cases}$$

Check: $\rho(0) = \rho_0$; as $t \to T_s^-$ the exponent tends to $-\infty$ so $\rho \to \rho_T$. Parameter roles: ρ_0 (initial bound, overshoot allowance), ρ_T (ultimate bound), T_s (settling time), κ (decay shape).

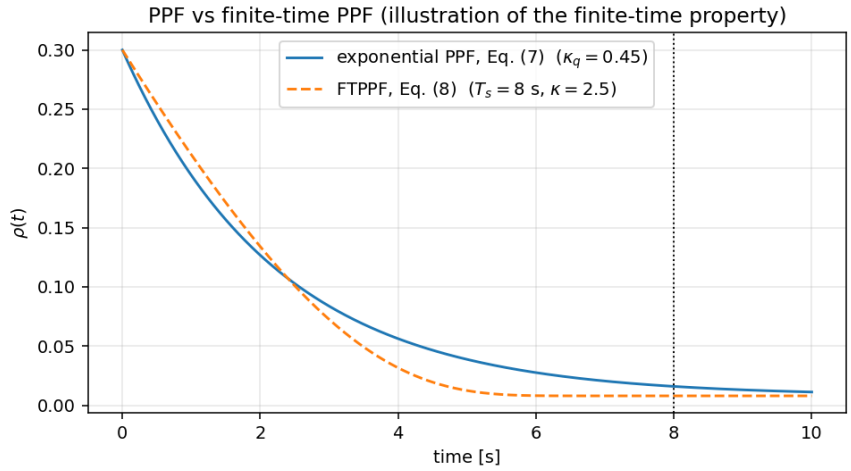


Figure 1. Illustration of the finite-time property: the exponential PPF (Eq. 7) is still above its limit at $T_s = 8$ s, whereas the FTPPF (Eq. 8) has reached ρ_T and stays there (same idea as the paper's Fig. 2).

2.3 Normalised errors (Eq. 11)

$$\varepsilon_1 = \frac{q_v}{\rho_q(t)}, \qquad \varepsilon_2 = \frac{\omega}{\rho_\omega(t)}$$

The constraint is satisfied if and only if every component satisfies $|\varepsilon_{li}| < 1$.

2.4 Constrained PD-type control law and adaptive law (Eq. 12–14)

$$u = -k_p\, q_v - k_d\, \omega - u_{cons}$$

$$u_{cons} = \left[\alpha\,(1-\varepsilon_1)^{-2} + \beta\,(1-\varepsilon_2)^{-2}\right] s + \hat{\theta}\,\mathrm{sgn}(s), \qquad s = \omega + c\,q_v,\ c > 0$$

$$\dot{\hat{\theta}} = \frac{1}{\eta_1}\Big(\|s\| - \eta_2\,\hat{\theta}\Big), \qquad \eta_1 > 0, \quad \int_0^\infty \eta_2(\tau)\,d\tau = \bar{\eta} < \infty$$

The bracket is applied component-wise (a diagonal gain matrix). To avoid chattering, Remark 5 (Eq. 22) replaces $\mathrm{sgn}(s)$ by $\mathrm{sat}(s/\varsigma)$.

Mechanism (Remark 2 and 7). When a state approaches its envelope, $\varepsilon \to 1$, the factor $(1-\varepsilon)^{-2}$ grows, the gain on $s = \omega + c\,q_v$ grows with it, and the control pushes the state back. $\hat{\theta}$ grows while $\|s\|$ is non-zero, which injects extra authority against the disturbance and saturation shortfall; the leakage $\eta_2(t)$ stops unbounded growth.

2.5 Stability result (Theorem 1)

Lyapunov candidate (Eq. 15; the paper denotes the scalar quaternion part by q_0 in this equation, written q_4 here), with $\tilde{\theta} = \theta - \hat{\theta}$:

$$V = (k_p + c\,k_d)(q_v^\top q_v + (1-q_4)^2) + \tfrac{1}{2}\,\omega^\top J\omega + c\,q_v^\top J\omega + \frac{1}{\eta_1}\tilde{\theta}^2$$

Using $\tilde{\theta}\hat{\theta} \le -\frac{2\sigma-1}{2\sigma}\tilde{\theta}^2 + \frac{\sigma}{2}\theta^2$ (Eq. 18), the derivative satisfies (Eq. 19)

$$\dot{V} \le -\lambda_{\min}(Q)\,\|x\|^2 + \frac{\eta_2\,\sigma}{2}\,\theta^2, \qquad x = \big[\|q_v\|,\ \|\omega\|,\ \tilde{\theta}\big]^\top$$

with $Q = \mathrm{diag}\big(c\,k_p,\ k_d - c\,\lambda_{\max}(J),\ \eta_2(2\sigma-1)/2\sigma\big)$. Positive-definiteness needs c small: $c < k_d/\lambda_{\max}(J)$. Integrating and applying Barbalat's lemma gives $q_v \to 0$, $\omega \to 0$.

Check on the paper's own numbers. For the paper's J, $\lambda(J) = \{14.27,\ 17.00,\ 20.74\}$ kg·m². With $k_d = 20$ the condition is $c < 0.964$. The paper does **not** state c; we use $c = 0.5$ (satisfies the condition, and also $2(k_p + c\,k_d)\lambda_{\min}(J) = 1427 > (c\lambda_{\max})^2 = 107$, so $P \succ 0$).

> **Important reading of Theorem 1.** As stated, Theorem 1 proves *convergence to the origin*. We could not find a separate formal proof that $|\varepsilon| < 1$ is preserved for all time. Constraint satisfaction is supported by Remarks 2 and 7 and by the simulations. This is relevant to how strongly we phrase our own claims (§8).

2.6 Paper's simulation set-up (Section IV)

Item	Value in paper
Inertia J	$\begin{bmatrix} 20 & 1.2 & 0.9 \\ 1.2 & 17 & 1.4 \\ 0.9 & 1.4 & 15 \end{bmatrix}$ kg·m²
Disturbance $d(t)$	$[0.05\sin t,\ 0.1\sin 1.2t,\ 0.15\sin 1.5t]^\top$ N·m
Actuator limit	$u_{\max} = 2.5$ N·m per axis
Gains	$k_p = 40,\ k_d = 20,\ \alpha = 1.5,\ \beta = 1.2$
Adaptive law	$\eta_1 = 0.1,\ \eta_2 = 2e^{-0.5t},\ \hat{\theta}(0) = 0$
FTPPF	$\rho_{q0} = 0.3,\ \rho_{\omega 0} = 0.4,\ \rho_{q\infty} = 0.008,\ \rho_{\omega\infty} = 0.005,\ \kappa = 2.5$
Part 1	$q(0) = [0.2, -0.15, -0.25, 0.9354]^\top$, $\omega(0) = [0.03, 0.02, -0.01]^\top$, settling time 15 s and 10 s
Part 2	$q(0) = [0.3, -0.2, -0.3, 0.8832]^\top$, $\omega(0) = [-0.02, -0.02, 0.025]^\top$, constrained vs conventional PD
Not specified	c, ς, settling time of Part 2, numerical solver and step

3. What Has Been Implemented

3.1 Repository contents

Component	Files	What it does	Maturity
A. Browser simulator	`index.html`, `app.js`, `simulation.js`, `style.css`	Interactive rigid-body attitude simulation, RK4 solver, constrained vs conventional PD, plots of quaternion, angular velocity, torque and $\hat{\theta}$	Working; barrier form differs from paper (F1)
B. MuJoCo attitude experiments	`Drones_Mujoco_Pointing_Trial.ipynb`	Direct ODE simulation of Eq. (1)–(14), side-by-side constrained vs PD bodies in a 3D viewer, arbitrary target orientation, point-the-nose-at-a-position	Strongest existing material; uses the paper's barrier form
C. MuJoCo flight experiments	`Drones_Mujoco_Flying.ipynb`, `Drones_Mujoco_Good_Trial.ipynb`	Attitude → altitude → point-to-point motion with outer position PD loop; minimum-jerk reference $(0,0,1) \to (3,3,3)$, final position about $(2.99985,\ 2.99985,\ 3.0)$	Working; simplified direct force/torque actuators; **not** evidence of the paper's controller flying

Component	Files	What it does	Maturity
D. Gazebo X3 controller	`gazebo.py`	ROS 2 node: pose from `/world/quadcopter/dynamic_pose/info`, FTPPF, barrier law, adaptive $\hat{\theta}$, altitude PD, tilt compensation, 4-motor mixer, publishes `/X3/gazebo/command/motor_speed`, logs and plots	Pipeline works; closed-loop hover **not yet sustained** (vehicle takes off, then loses attitude control and crashes)
E. Webots script	`simulation_for_webots.py`	Motor-control experiment with plots	Conventional-PD curves are *synthetically modified*, not independently simulated (F6)
F. Early script	`simulation.py`	Earlier controller draft	File begins with a shell heredoc (`cat << 'EOF'`), so it is not valid Python as stored (F5)
Paper and media	`base paper.pdf`, images, two videos	Reference paper and demo recordings	—

3.2 Gazebo X3 pipeline (component D) in detail

State acquisition: z and the quaternion are read from the simulator pose message; the quaternion sign is kept continuous. Because only pose is used, ω and $\dot{z}$ are **estimated** by backward differences and first-order low-pass filtering:

$$\omega_{raw} \approx \frac{2\,(q_{v,k} - q_{v,k-1})}{\Delta t}, \qquad \omega_k = (1-\lambda_\omega)\,\omega_{k-1} + \lambda_\omega\,\omega_{raw}, \;\; \lambda_\omega = 0.10$$

Altitude and thrust:

$$F_z = m\Big(g + \mathrm{clip}\big(K_{p,z}(z_d - z) - K_{d,z}\dot{z},\ \pm 3\big)\Big), \qquad T = \mathrm{clip}\Big(\frac{F_z}{\mathrm{clip}(\cos 2q_1 \cos 2q_2,\ 0.85,\ 1)},\ 0,\ 22\Big)$$

with $m = 1.5$ kg, $K_{p,z} = 5$, $K_{d,z} = 3.5$, target altitude 2 m. Attitude torques $\tau = [\tau_1, \tau_2, \tau_3]^\top$ come from the constrained law and are mapped to rotor speeds by the standard mixer (arm $l = 0.225$ m, $c_T = 8.54858 \times 10^{-6}$, $c_Q = 0.016$):

$$\omega_{rotor,i} = \sqrt{\max(0,\ m_i)}, \qquad m_{1,4} = \frac{T}{4c_T} \pm \frac{\tau_2}{4c_T l} \mp \frac{\tau_1}{4c_T l} \mp \frac{\tau_3}{4c_Q}$$

3.3 Stable trajectory baseline (separate)

A separate Python node generates a 1.5 m square at 1.5 m altitude, computes position error and sends velocity commands to Gazebo's built-in multicopter velocity controller. Plots exist for desired vs actual 3D path, per-axis tracking and position error. It validates the simulation pipeline and the evaluation tooling. **Action:** this script and its CSV/plots were not in the repository snapshot reviewed for this report; commit them (under `baseline/`) before the review.

4. New in This Report: Faithful Reproduction of the Paper's Simulations

`paper_reproduction.py` implements Eq. (1)–(14) exactly (RK4, $\Delta t = 1$ ms, zero-order-hold torque, saturation Eq. 3–4, boundary-layer form Eq. 22) and reproduces both experiments of the paper's Section IV. It writes every figure and number below to `docs/figures/` and `docs/results/`. Parameters the paper does not state are listed in §2.6 and chosen as $c = 0.5$, $\varsigma = 0.02$.

Run it: `python paper_reproduction.py` (needs numpy and matplotlib).

4.1 Part 1: effect of the FTPPF settling time

Metric	$T_s = 15$ s	$T_s = 10$ s
Max normalised error, $\max\|\varepsilon_1\|$ / $\max\|\varepsilon_2\|$	0.871 / 0.912	0.924 / 0.941
Time outside envelope, q_v / ω	0 % / 0 %	0 % / 0 %
Settling time ($\|q_v\| < 0.01$)	5.25 s	4.53 s
Final $\|q_v\|$ / $\|\omega\|$	4.4×10^{-4} / 4.0×10^{-3} rad/s	4.5×10^{-4} / 4.1×10^{-3} rad/s
Time with commanded torque at limit	6.5 %	5.9 %
$\hat{\theta}$ at 20 s	0.433	0.416

Result. Both states stay strictly inside their envelopes ($\max|\varepsilon| < 1$) and a smaller T_s gives faster convergence, matching the paper's Part 1 conclusion.

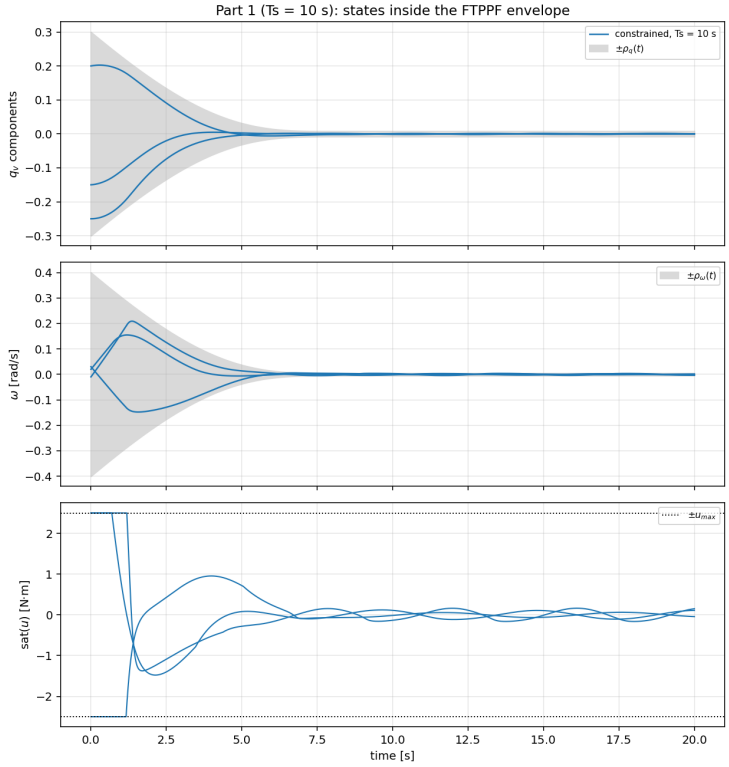


Figure 2. Part 1 ($T_s = 10$ s). Grey region is $\pm\rho(t)$; the states never touch it. Bottom panel is the applied torque, saturated at 2.5 N·m.

4.2 Part 2: constrained PD-type vs conventional PD (same gains, saturation, disturbance, initial state)

Metric	Constrained (paper Eq. 13)	Constrained (symmetric barrier)	Conventional PD
$\max\|\varepsilon_1\|$	0.868	0.868	0.868
$\max\|\varepsilon_2\|$	0.904	0.837	**1.496**
Time outside q_v envelope	0 %	0 %	0 %
Time outside ω envelope	**0 %**	**0 %**	**27.5 %**
Time commanded torque at limit	6.3 %	7.2 %	14.0 %
Peak commanded torque (before saturation)	18.8 N·m	13.8 N·m	11.6 N·m
Settling time ($\|q_v\| < 0.01$)	5.79 s	4.19 s	6.31 s
Final $\|\omega\|$	0.0040 rad/s	0.0040 rad/s	0.0083 rad/s
Torque energy $\int\|\text{sat}(u)\|^2 dt$	38.9	41.5	60.5
Final $\hat{\theta}$	0.439	0.435	n/a

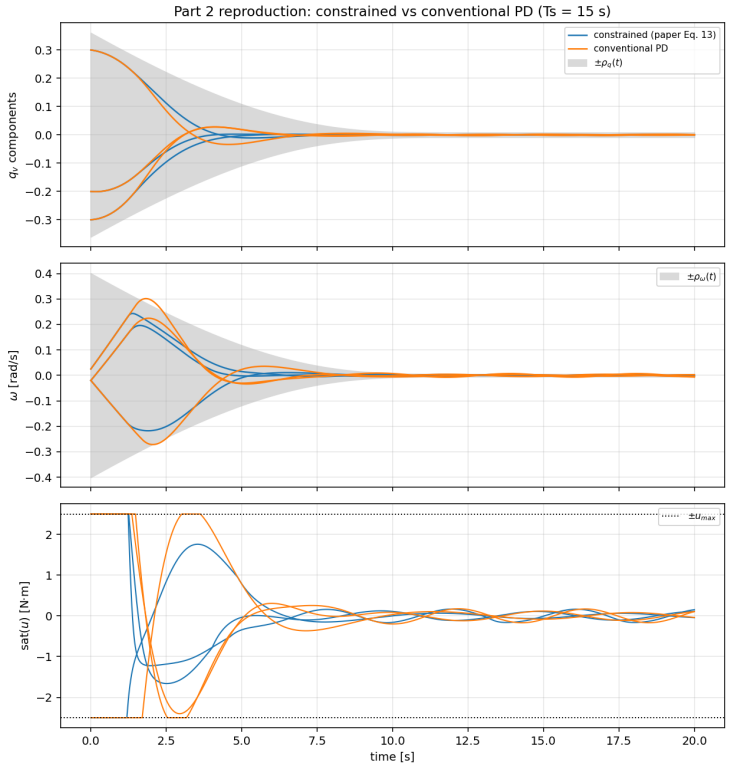


Figure 3. Part 2 reproduction ($T_s = 15$ s, $\rho_{q0} = 0.36$). Both controllers use the same 2.5 N·m limit. The conventional PD leaves the angular-velocity envelope; the constrained law does not.

Result. The paper's central claim is reproduced qualitatively: with the same actuator limit, the constrained controller keeps ω inside its envelope (0 % violation versus 27.5 % for PD), reaches a smaller final angular velocity, uses about 36 % less torque energy, and spends less time at the torque limit. The paper itself publishes plots only, so the comparison is qualitative, not number-for-number.

Note. The conventional PD happens to settle the quaternion at a similar time and the conventional PD's quaternion also stays inside its envelope; the advantage is mainly in the **angular-velocity envelope**, the **final residual** and the **control effort**. The browser and MuJoCo notebook runs, which print only $\|q_v\|$ versus time, therefore do not by themselves show a clear win for the constrained law (at some times the PD value is smaller). The metrics above are the right way to present the comparison.

4.3 Sensitivity to the parameters the paper does not specify

Case (constrained, Part 2)	q outside	ω outside	Torque at limit	$\max\lvert\varepsilon_1\rvert$
$T_s = 15$ s, $\rho_{q0} = 0.36$ (main)	0 %	0 %	6.3 %	0.87
$T_s = 15$ s, $\rho_{q0} = 0.50$	0 %	0 %	7.7 %	0.63
$T_s = 10$ s, $\rho_{q0} = 0.50$	0 %	9.5 %	7.4 %	0.81
$T_s = 10$ s, $\rho_{q0} = 0.36$	**65.9 %**	**55.9 %**	**71.2 %**	**6.55**

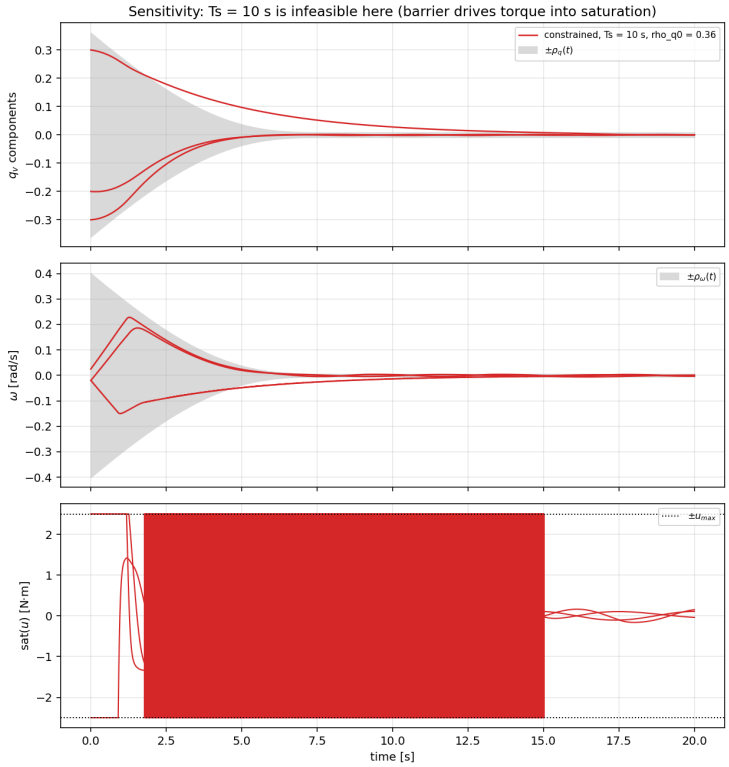


Figure 4. With $T_s = 10$ s and a tight ρ_{q0} the envelope demands more than the 2.5 N·m actuator can deliver; the barrier gain saturates the torque and the constraint is lost.

Lesson. The constraint guarantee is only meaningful when the envelope is **feasible for the actuator**: T_s, ρ_0, ρ_T and $u_{\max}$ cannot be chosen independently. This is directly relevant to the Gazebo X3, whose actuator authority is very different from the paper's spacecraft (§6).

4.4 Adaptive estimate

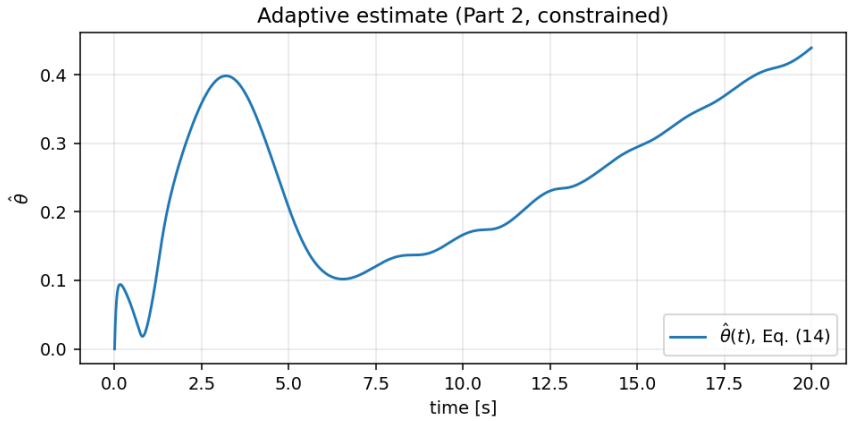


Figure 5. $\hat{\theta}$ grows while $\|s\|$ is non-zero and levels off through the leakage $\eta_2(t)$, as designed.

5. Equation-to-Code Audit

5.1 Traceability matrix

Paper equation	`gazebo.py`	`simulation.js`	MuJoCo pointing notebook	Reproduction script
Plant (1), (5)	not simulated; real Gazebo physics	`rigidBodyDynamics` + RK4	direct ODE / MuJoCo body	`simulate()`
Saturation (3)–(4)	clip to $\pm u_{\max}$ (l. 732)	clip	`saturate()`	`sat()`
FTPPF (8)	`ftppf()` (l. 226)	`ftppf()` (l. 61)	`ftppf()`	`ftppf()`
$\varepsilon_{1,2}$ (11)	l. 577; clipped to ± 0.9	l. 151–171; clipped to ± 0.9999	no clipping	clipped to ± 0.999 (guard only)
Barrier in (13)	l. 611: $\alpha/(1-\varepsilon^2)^2$	l. 173: $\alpha/(1-\varepsilon^2)^2$	$(1-\varepsilon)^{-2}$ (**as paper**)	both forms selectable
u_{cons} (13), (22)	l. 627; then clipped to ± 0.05 (l. 650)	l. 176	`coeff*s + theta_hat*sat(s/zeta)`	as paper
Control law (12)	l. 710	l. 180	`constrained_controller()`	`control()`
Adaptive law (14)	l. 671; $\hat{\theta}$ clipped to $[0, 1]$ (l. 695)	l. 185	Euler update	Euler update, $\hat{\theta} \geq 0$

5.2 Findings

#	Finding	Evidence	Impact	Action
F1	**Barrier form differs from the paper.** Paper Eq. 13 uses $(1-\varepsilon)^{-2}$. `gazebo.py`, `simulation.js`, and `simulation_for_webots.py` use $(1-\varepsilon^2)^{-2}$. The MuJoCo pointing notebook matches the paper.	Fig. 6, code lines above	The two forms are not the same controller. The paper's form is *one-sided* (it only grows as $\varepsilon \to +1$; at $\varepsilon \to -1$ it equals $1/4$). The symmetric form is two-sided.	Treat the paper form as the **reference**; keep the symmetric form as a documented **variant**; label which one each result uses. In our reproduction both keep Part 2 inside the envelope (§4.2).

#	Finding	Evidence	Impact	Action
F2	**Paper's Part 2 initial condition lies on the boundary.** $\lvert q_1(0)\rvert = \lvert q_3(0)\rvert = 0.3 = \rho_{q0}$, so $\varepsilon_1(0) = 1$ and $(1-\varepsilon_1)^{-2}$ is infinite. This violates the paper's own Remark 7 ($\lvert q_i(0)\rvert < \rho_{q0}$).	MuJoCo notebook prints `divide by zero encountered in power`	The paper must have used a larger ρ_{q0} for Part 2 that it does not state.	Use $\rho_{q0} = 1.2\max\lvert q_i(0)\rvert = 0.36$ (as the notebook does) and document it. Done in the reproduction.
F3	**Gazebo envelope is almost inactive.** $\rho_{q0} = 2.0$ (a unit-quaternion component can never exceed 1, so $\varepsilon_1 \le 0.5$ always), $\rho_{qT} = 0.30$ (paper: 0.008), $\rho_{\omega T} = 3$ rad/s (paper: 0.005), $\alpha = \beta = 0.01$ (paper 1.5, 1.2), plus u_{cons} clipped to ± 0.05 N·m. The barrier gain at $\varepsilon = 0,\ 0.5,\ 0.9$ is only $0.01,\ 0.018,\ 0.277$ (paper form with $\alpha = 1.5$: 1.5, 6, 150).	`gazebo.py` l. 93–113, 650; `docs/results/metrics.json`	In practice the Gazebo controller behaves like a **plain PD** ($k_p = 1.5,\ k_d = 0.6$) plus a small term. Hence the barrier is unlikely to be the *cause* of the instability, and it also cannot be credited with providing constraints.	Experiment E1 (§7).
F4	**Gazebo ω is estimated, not measured.** Finite difference of q_v plus a 0.10 low-pass filter introduces lag and noise exactly where the PD term ($k_d\omega$) and $s = \omega + c\,q_v$ need a clean signal.	`gazebo.py` l. 429–440	A plausible instability contributor (hypothesis H2).	Experiment E2.
F5	`simulation.py` is not runnable Python (starts with `cat << 'EOF' > drone_brain.py`).	line 1	Dead file in the repo.	Strip the heredoc wrapper or delete the file.
F6	`simulation_for_webots.py` builds the conventional-PD curves by *modelled* modification ("Benchmark modeling: Conventional PD has residual overshoot…"), not by a second controlled simulation.	comment near l. 293	Its comparison plots are illustrative, not a benchmark.	Replace by `paper_reproduction.py` results or fix the script to simulate PD independently.
F7	$\hat{\theta}$ is hard-clipped to $[0, 1]$ in Gazebo, and the sign function is replaced by a boundary layer.	`gazebo.py` l. 695	Departures from Eq. (14) / (13) that are reasonable engineering guards but must be disclosed.	Document as implementation guards.
F8	**Theorem 1 proves convergence, not invariance of the envelope.**	paper §III-C	Wording of our claims.	Phrase as "constraint satisfaction observed in simulation".

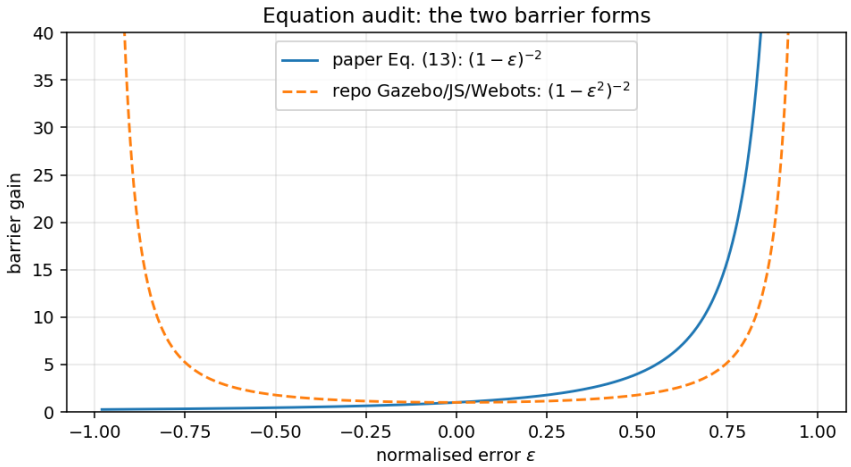


Figure 6. The two barrier forms in use. The paper's form is one-sided; the repository's Gazebo/JS/Webots form is symmetric.

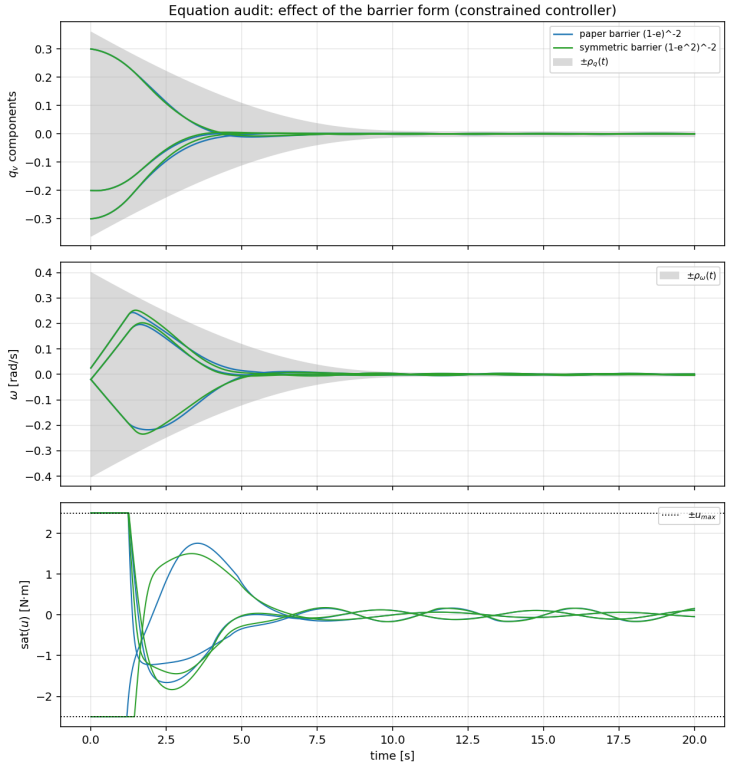


Figure 7. Effect of the barrier form in Part 2. Both keep every state inside the envelope; the symmetric form reaches a lower peak $|\varepsilon_2|$ (0.837 vs 0.904).

6. Current Problem: Paper Controller on the Gazebo X3

Observed behaviour. In the Gazebo runs executed so far, the X3 takes off and climbs under the controller. Attitude control is not sustained: after the ascent the vehicle loses stability, descends in an uncontrolled manner and the run terminates in a crash. The simulation pipeline itself (pose acquisition, motor-speed commands, logging and plotting) operates correctly, so the problem is confined to the closed-loop controller and its parameters.

Status of the cause. The root cause has not yet been established. Each item below is a **hypothesis** to be tested by a controlled experiment, not a conclusion.

6.1 Parameter gap between the paper and the X3

Quantity	Paper (spacecraft)	`gazebo.py` (X3)
Inertia	$\approx$ 14–21 kg·m²	to be read from the X3 SDF (expected order 10^{-2} kg·m²)
Torque limit	2.5 N·m	0.20 N·m (then limited again to about 0.08–0.20 N·m by the mixer headroom rule)
$k_p,\ k_d$	40, 20	1.5, 0.6
$\alpha,\ \beta$	1.5, 1.2	0.01, 0.01
$\rho_{q0},\ \rho_{qT}$	0.3, 0.008	2.0, 0.30
$\rho_{\omega 0},\ \rho_{\omega T}$	0.4, 0.005	10, 3
T_s	10–15 s	10 s
Angular velocity	state (assumed measured)	finite-difference estimate

Gain-scaling rule we will use. For small angles $q_v \approx \theta/2$, so the inner loop is $J\ddot{\theta} = -\frac{k_p}{2}\theta - k_d\dot{\theta}$, giving

$$\omega_n = \sqrt{\frac{k_p}{2J}}, \qquad \zeta = \frac{k_d}{2J\omega_n}$$

For the paper's data ($J \approx 17$) this is $\omega_n \approx 1.07$ rad/s, $\zeta \approx 0.54$. We will choose X3 gains to obtain a comparable ω_n, ζ from the X3's *actual* inertia, and then choose T_s, ρ_0, ρ_T so the envelope is feasible for the X3's actuator (lesson of §4.3).

6.2 Hypotheses and the experiment that tests each

ID	Hypothesis	Experiment (single controlled change)	Evidence that confirms	Cost
H5	The barrier term is not the cause (F3)	Run the existing `controller_mode='conventional_pd'` under identical conditions	PD alone also fails → the cause lies elsewhere	Very low (flag exists)
H2	Finite-difference ω lag/noise destabilises the loop	Log estimated ω against the simulator's true angular velocity (IMU or twist topic); then close the loop on the true value	Phase lag/noise visible; stability improves with true ω	Low
H3	A sign or frame error in the torque→motor mapping	Open-loop test: apply a small step torque per axis and compare the measured response direction	Wrong sign on at least one axis	Low
H4	Mixer or thrust headroom saturates motors	Log the four m_i and the fraction clipped at 0 or maximum; vary the headroom rule	Clipping coincides with divergence	Low
H1	Gains inconsistent with the X3 inertia and actuator authority	Read inertia from SDF, apply the ω_n, ζ rule, retune	Stable response at the designed ω_n, ζ	Medium

ID	Hypothesis	Experiment (single controlled change)	Evidence that confirms	Cost
H6	Time step jitter in the pose callback (Δt is taken from message timing)	Log Δt statistics; use simulator time	Large Δt variance	Low

7. Plan for the Next Review

7.1 Objective

> **Validate the paper's constrained PD-type controller against its source equations in the mathematical simulator, quantify its benefit over conventional PD, and demonstrate evidence-based progress on the Gazebo X3 integration.**

We deliberately do **not** promise a stable Gazebo flight with the paper controller. We promise a specific, evidenced investigation, and a stable flight if the evidence allows.

7.2 Work packages and acceptance criteria

WP	Task	Deliverable	Acceptance criterion	Status
WP1	Equation audit and alignment	Table 5.1; one agreed reference form (paper barrier) and a documented variant; all files labelled	Every implementation either matches Eq. (12)–(14) or declares its deviation	Audit **done**; alignment of JS/Gazebo/Webots **to do**
WP2	Quantitative reproduction of Part 1 and Part 2	`paper_reproduction.py`, metrics tables, figures	$\lvert\varepsilon\rvert < 1$ throughout for the constrained law; PD comparison with the metrics of §7.3	**Done** (this report)
WP2b	Robustness study (new)	Monte-Carlo over initial conditions inside the envelope and over disturbance amplitude; success-rate table	Violation-free fraction reported for constrained and PD with confidence interval	To do
WP3	Gazebo diagnosis: experiments H5 → H2 → H3 → H4 → H1	One notebook/log per experiment; one-page findings table	At least one hypothesis confirmed or rejected with logged evidence; one justified change applied	To do
WP4	Gazebo benchmark, *only if WP3 yields a stable hover*	Constrained vs conventional PD on the X3, with and without the paper's disturbance, same metrics	60 s stable hover in both modes with logged metrics	Conditional
WP5	Stable trajectory baseline packaged	`baseline/` script, CSV, desired-vs-actual 3D plot, per-axis plots, RMS position error	Reproducible from a clean checkout in one command	Done, needs committing
WP6	Code consolidation	Repair or remove `simulation.py`; relabel the Webots comparison as illustrative; label each file with the barrier form it uses	Every file in the repository is runnable and labelled accurately	To do

7.3 Metrics (defined once, used everywhere)

With T the simulation length and $\mathbf{1}[\cdot]$ the indicator:

$$V_q = \frac{100}{T}\int_0^T \mathbf{1}\big[\exists i:\ |q_i| \geq \rho_q\big]\,dt, \qquad V_\omega = \frac{100}{T}\int_0^T \mathbf{1}\big[\exists i:\ |\omega_i| \geq \rho_\omega\big]\,dt$$

$$\varepsilon_{\max} = \max_{t,i,l} |\varepsilon_{li}|, \qquad S_u = \frac{100}{T}\int_0^T \mathbf{1}\big[\exists i:\ |u_i| \geq u_{\max}\big]\,dt, \qquad E_u = \int_0^T \|\mathrm{sat}(u)\|^2 dt$$

plus settling time (last instant $\|q_v\| > 0.01$), final $\|q_v\|$ and final $\|\omega\|$. For the Gazebo trajectory baseline: RMS position error $\sqrt{\frac{1}{N}\sum\|p_k - p_k^d\|^2}$ and maximum error.

7.4 What will be shown at the next review

Slide / demo	Content	Source
1. Paper and equations	Problem, FTPPF, ε-normalisation, control law, adaptive law, Lyapunov conditions (§1–2)	this report
2. Equation audit	Traceability matrix and findings F1–F3, with Fig. 6–7	§5
3. Reproduction	Part 1 and Part 2 figures, metrics table, sensitivity (§4)	`paper_reproduction.py`
4. MuJoCo demo	Live attitude-pointing, constrained vs PD side by side	`Drones_Mujoco_Pointing_Trial.ipynb`
5. Gazebo X3	Architecture diagram with actual topics, the diagnostic experiments run, their evidence, and the stable baseline plots, clearly labelled as a *different* controller	`gazebo.py`, `baseline/`
6. Limits and next steps	What is still unproven and why	§6, §8

7.5 Schedule for the coming week

Days	Focus	Exit condition
1	Commit the trajectory baseline (WP5); consolidate code (WP6); align barrier form across files (WP1)	Repository consistent with this report
2–3	Gazebo experiments H5, H2 and H3 (WP3)	Logged result and one-line conclusion for each
4	Experiments H4 and H1; apply one justified change (WP3)	Gazebo result: stable hover, or a documented failure mode
5	Robustness study in the mathematical simulator (WP2b)	Table of violation-free fractions
6	Slides and demo preparation	Demo runs from a clean checkout
7	Rehearsal and buffer	Review-ready

7.6 Risks

Risk	Likelihood	Mitigation
Gazebo hover not achieved within the week	Medium	The deliverable is the *diagnosis*; MuJoCo and reproduction results carry the technical content
Envelope infeasible for X3 actuator limits (cf. §4.3)	Medium	Choose T_s, ρ_0, ρ_T from the X3 torque authority; report the feasibility check
Reviewer questions the one-sided barrier or Theorem 1 scope	Medium	F1/F8 already analysed; ablation in Fig. 7
Unstated paper parameters (c, ς, Part 2 T_s)	Certain	Disclosed in §2.6; sensitivity in §4.3

8. Scope of Claims

Claims supported by the present work:

- The paper's equations (1)–(14) are implemented and reproduce its Part 1 and Part 2 behaviour qualitatively. In Part 2, at the same torque limit, the constrained law keeps ω inside its envelope throughout, while conventional PD leaves it for 27.5 % of the time.
- The FTPPF, normalised-error, adaptive-law and saturation components are implemented in the MuJoCo, browser and Gazebo code.
- The ROS 2–Gazebo X3 pipeline (pose input, motor-speed output, logging and plotting) is operational, and the X3 takes off and climbs under the implemented controller.
- A square-trajectory baseline using Gazebo's built-in velocity controller has been demonstrated and is kept separate from the paper controller.

Claims deferred until supported by further results:

- Sustained hover of the X3 under the paper controller.
- Use of the paper's adaptive controller in the square-trajectory baseline (it uses Gazebo's built-in low-level controller).
- A four-rotor model in the MuJoCo point-to-point flights (they use direct force and torque actuators).
- An independent constrained-versus-conventional benchmark in Webots.
- A proof that the envelope is preserved for all time (the paper's Theorem 1 establishes convergence only).

Status statement for this review:

> *The constrained PD-type controller has been implemented and validated in a direct numerical simulation of the paper's rigid-body model, where it keeps both the quaternion and the angular velocity inside the finite-time envelopes at a 2.5 N·m actuator limit. Integration with the Gazebo X3 quadcopter is in progress: the vehicle takes off and climbs, but a sustained hover has not yet been achieved, and a diagnostic study is planned for the coming week.*

9. Reproducing the Results

```
pip install numpy matplotlib
python paper_reproduction.py        # writes docs/figures/*.png and
 ↪  docs/results/metrics.{json,md}
```

Output	File
Part 1 figures and bounds	`docs/figures/fig_part1.png`, `fig_part1_bounds.png`
Part 2 comparison	`docs/figures/fig_part2.png`, `fig_theta_hat.png`
Equation audit	`docs/figures/fig_barrier_forms.png`, `fig_barrier_ablation.png`
Feasibility sensitivity	`docs/figures/fig_part2_Ts10_infeasible.png`
All numbers	`docs/results/metrics.json`, `metrics.md`

10. References

1. M. Golestani *et al.*, "A Simple Structure Constrained Attitude Control for Rigid Bodies: A PD-Type Control," *IEEE Access*, vol. 10, pp. 10202–10209, 2022.
2. C. P. Bechlioulis and G. A. Rovithakis, "Robust adaptive control of feedback linearizable MIMO nonlinear systems with prescribed performance," *IEEE Trans. Autom. Control*, vol. 53, no. 9, pp. 2090–2099, 2008. (origin of PPC; paper ref. [22])
3. J. T.-Y. Wen and K. Kreutz-Delgado, "The attitude control problem," *IEEE Trans. Autom. Control*, vol. 36, no. 10, pp. 1148–1162, 1991. (Lyapunov function basis; paper ref. [42])
4. H. K. Khalil, *Nonlinear Systems*, Prentice-Hall, 2002. (Barbalat's lemma; paper ref. [43])